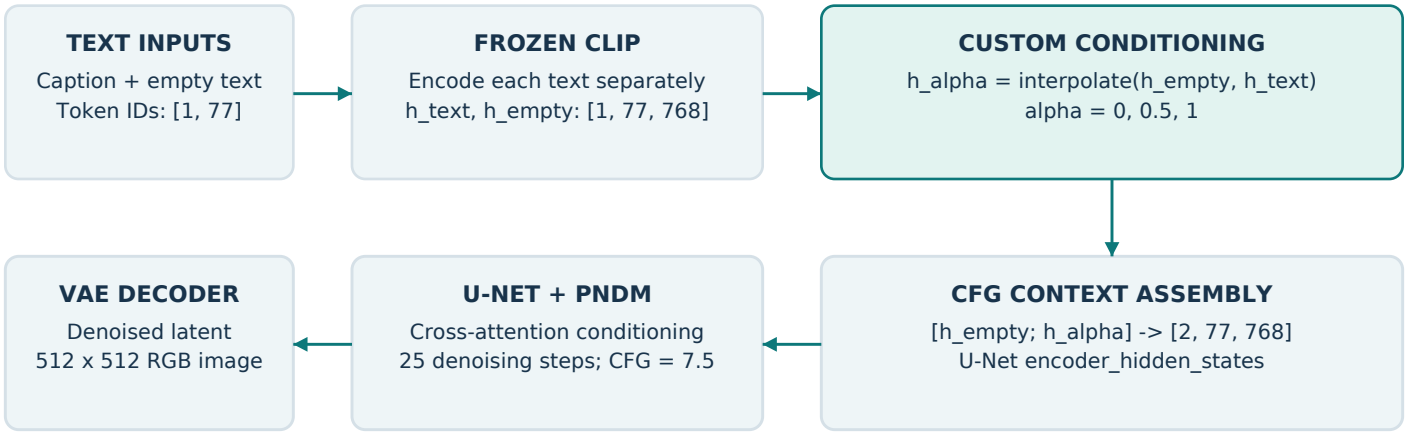
Text Encoder-Diffusion Integration

Project 2 - Milestone 2 | Point 1 integration write-up
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1. Scope and implementation

The project makes the CLIP-to-diffusion connection explicit and adds a controlled transformation of its conditioning embeddings. The implementation uses **stable-diffusion-v1-5/stable-diffusion-v1-5**. CLIP, U-Net and VAE weights remain frozen; no training or fine-tuning is performed. The contribution is an inference-time conditioning modification, a reusable generation function, and instrumented verification of the integration.

Architecture and observed tensor flow



Seeded noise: [1, 4, 64, 64]; duplicated U-Net input observed: [2, 4, 64, 64]

Figure 1. Text is encoded before diffusion, and the custom embeddings enter the U-Net as cross-attention context [1, 2]. A forward hook verified the actual conditioning tensor **[2, 77, 768]**. Its batch size of two represents the unconditional and conditional branches of classifier-free guidance (CFG).

2. Modification of the conditioning path

```
h_alpha = h_empty + alpha * (h_text - h_empty)
```

encode_text_explicitly() tokenizes each input independently and obtains CLIP token-level hidden states of shape **[1, 77, 768]**. These match the U-Net's 768-dimensional conditioning input. The 77 positions include special and padding tokens; these are not pooled 512-dimensional retrieval embeddings.

generate_with_conditioning() supplies the transformed tensor through **prompt_embeds**, retaining h_empty as **negative_prompt_embeds** [1]. Alpha 0 uses empty conditioning, alpha 0.5 interpolates in float32 before casting to float16, and alpha 1 passes the unmodified text embeddings directly. This changes the embedding input while keeping CFG scale fixed; it does not replace the denoiser or alter model weights.

Implementation note: an initial prototype encoded both texts in one batch and interpolated even at alpha 1. Its dog-prompt pixel MAE was 3.5974. The final implementation uses separate encoding and direct endpoints; its final compatibility results are reported on page 2. The individual cause of the earlier discrepancy was not isolated.

Verification and observed results

3. Experimental protocol

Four captions from the project's validation split were tested using the original pipeline and fixed alpha values 0, 0.5 and 1 (16 images; seeds 1001-1004). The 16 original baseline captions were tested only with the original pipeline and alpha 1 (32 images; seeds 42-57). This yields **48 images and 20 paired compatibility checks**. No setting was selected from evaluation outputs.

All runs used **512 x 512** images, **25** inference steps, **CFG 7.5**, PNDMScheduler and float16 on a Tesla T4. The same seed was reset within each comparison. Recorded versions: PyTorch 2.14.1+cu130, Diffusers 0.42.1 and Transformers 5.22.1.

FINAL VERIFICATION	MEASURED RESULT
Alpha 1 vs. original pipeline	20/20 pixel-identical pairs; mean and max MAE = 0
Generation time	206.13 s total; 4.29 s/image mean
Peak allocated GPU memory	3257.87 MiB (3.18 GiB)

Timing includes text encoding and pipeline processing, but excludes model loading and file writing. Memory is peak allocated process memory, including resident weights. Independent inspection of the archive also found matching PNG hashes for all 20 original/alpha-1 pairs.

4. Qualitative conditioning check

Validation v03, seed 1003: "A train sitting idle in an empty train station".

alpha = 0 **alpha = 0.5** **alpha = 1**

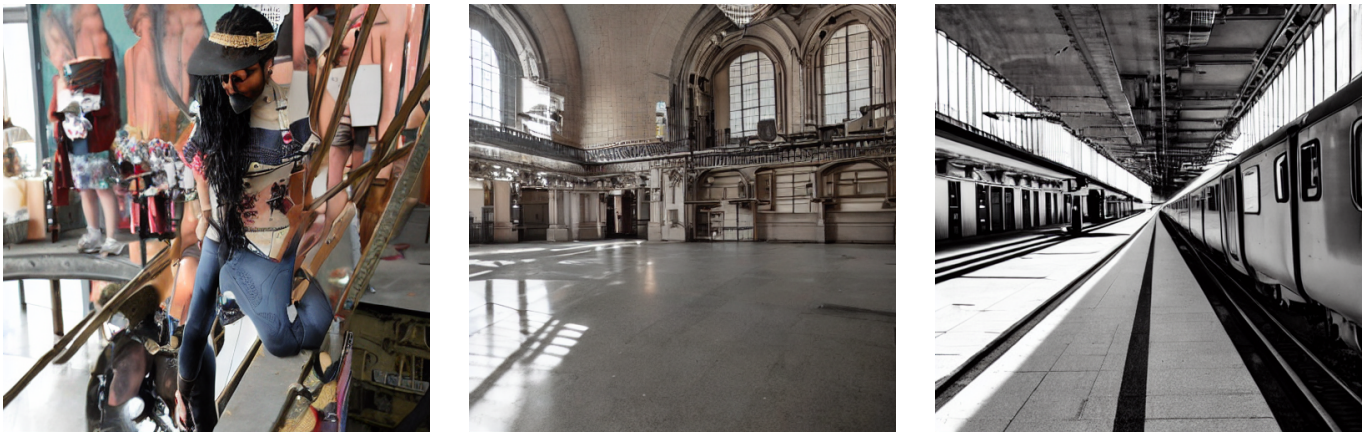


Figure 2. Alpha 0 produces a scene unrelated to the caption; alpha 0.5 produces a station-like interior without a visible train; alpha 1 produces trains at platforms. At alpha 0.5, v01 retains parked cars and v02 retains a person on pavement but changes the board geometry. In v04, the girls and doughnuts appear together at alpha 1. These are visual observations, not alignment scores.

5. Limitations and reproducibility

The tests verify baseline compatibility and sensitivity to the conditioning input, not a quality improvement. Each prompt uses one seed; no quantitative alignment or diversity metric was computed. Intermediate weights need not yield an intermediate semantic result. Same-session pixel equality does not guarantee equality across environments. Next steps are multi-seed validation and explicit prompt-alignment assessment.

Evidence: **05_text_encoder_integration.ipynb**; **run_logs.csv**; **compatibility_checks.csv**; **architecture_trace.json**; **environment.json**; **test_prompts.csv**; saved PNGs and galleries. Preserve these extracted artifacts with the notebook.

References

[1] Hugging Face. Diffusers: Stable Diffusion text-to-image pipeline. Official API documentation. Accessed November 9, 2026.

[2] Rombach et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. CVPR. arXiv:2112.10752.